Task-4 Implement NAT Gateway Setup

Description

i. Create two instances, one in a public subnet and one in a private subnet
ii. Host a website inside public web server
iii. Host a database inside private database server
iv. Connect to DB server only from its webserver, shouldn't have a public access to server even if the security group has open entries
v. DB Server should have internet access. Try to update the software packages, patching, etc

Step-1 Create a VPC

Custom-VPC	vpc-0e25e57ebdbf81dd0	Available	–	–

Step-2 Created two subnets Public and Private.

Subnets (6) Info

Last updated 5 minutes ago | Actions | Create subnet

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-0de88b6fef9338dc8	Available	vpc-0bbc07bb5ff3a0849	Off	172.31.32.0/20
Private Subnet	subnet-077332344d334fb29	Available	vpc-0e25e57ebdbf81dd0 | Cust...	Off	192.168.2.0/24
Public Subnet	subnet-08726592e0d6268a9	Available	vpc-0e25e57ebdbf81dd0 | Cust...	Off	192.168.1.0/24

Step-3 Created Internet gateway named IGW

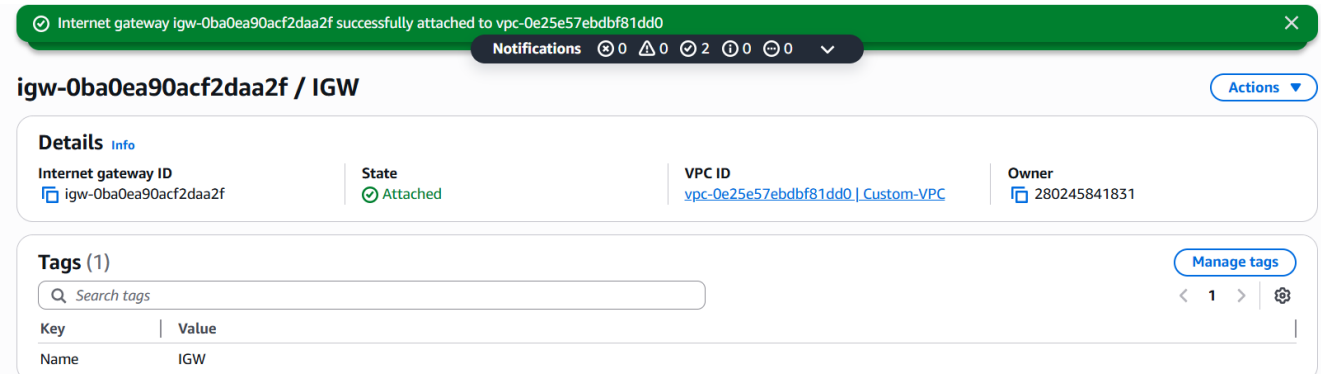
Internet gateways (1/3) Info

Actions | Create internet gateway

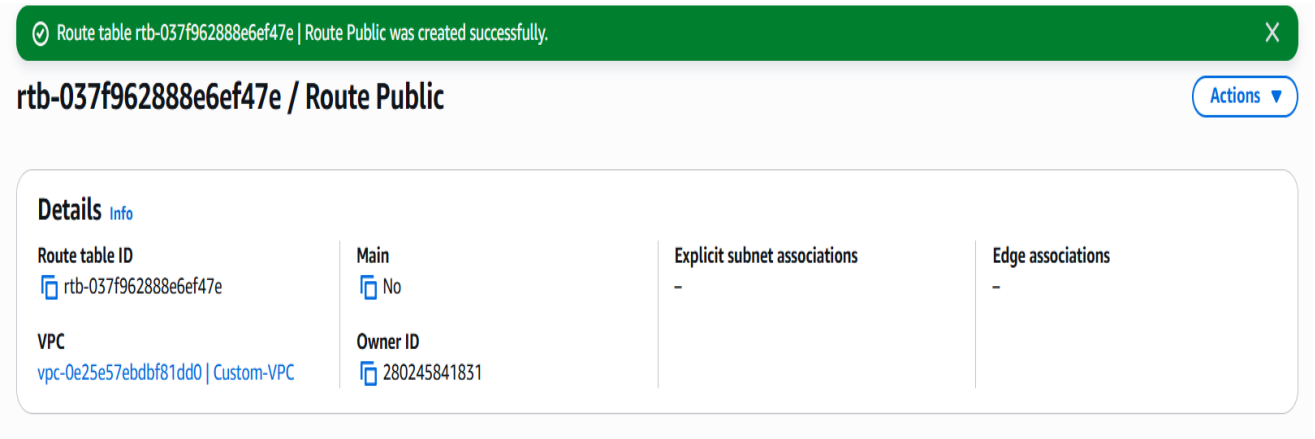
Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-04e00d1b4b5fdb2c2	Attached	vpc-0bbc07bb5ff3a0849	280245841831
my-internet-gateway	igw-0d8cc834bbee88ce1	Attached	vpc-0e9a839d62733855b | my-vpc	280245841831
IGW	igw-0ba0ea90acf2daa2f	Detached	-	280245841831

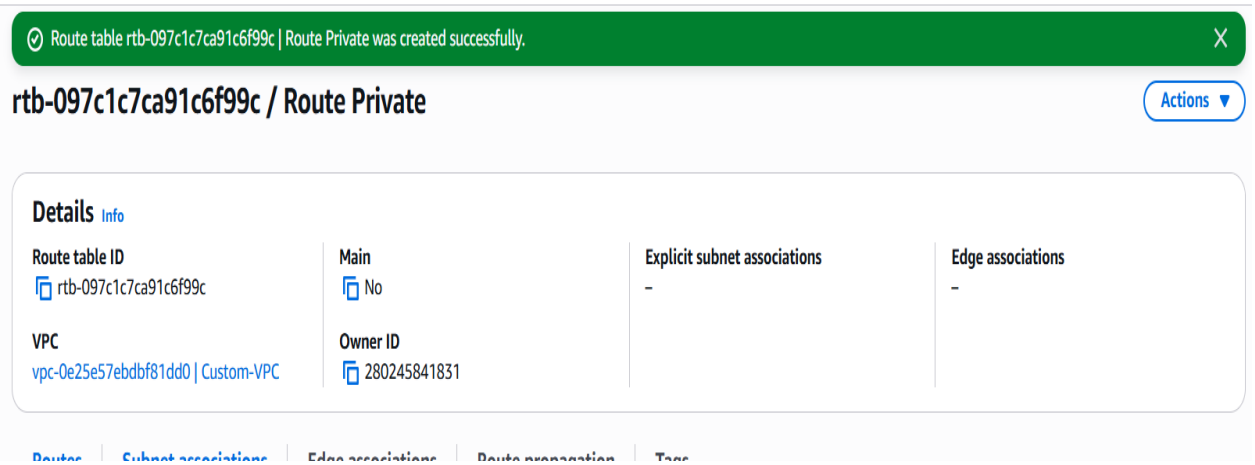
Step-4 Attached Internet gateway to VPC



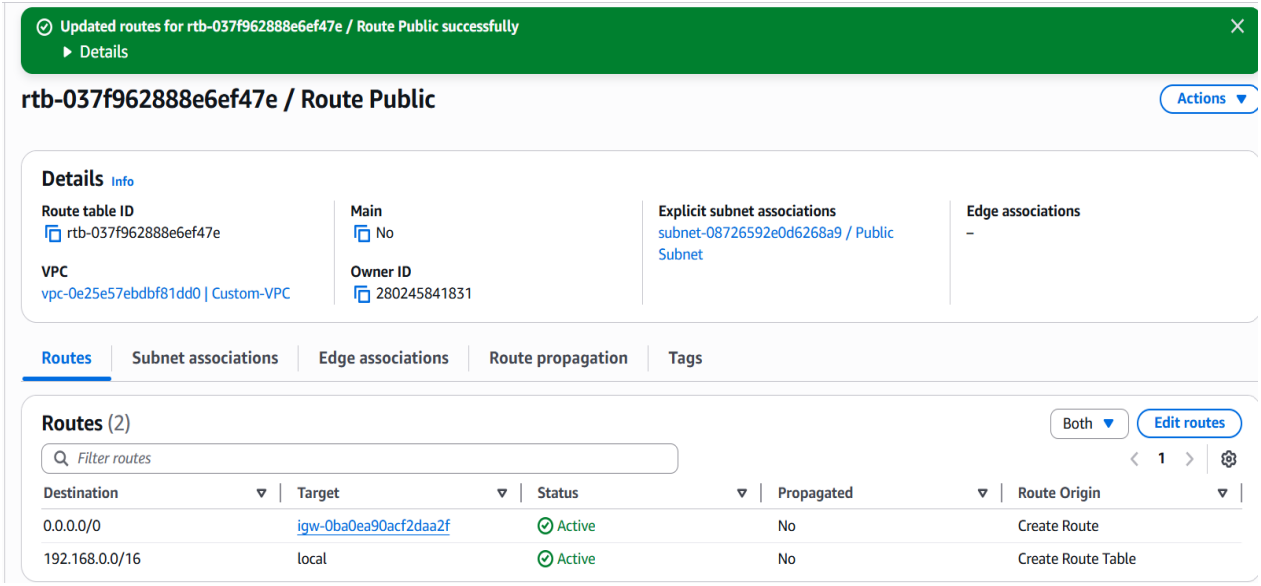
Step-5 Created Route Table Public



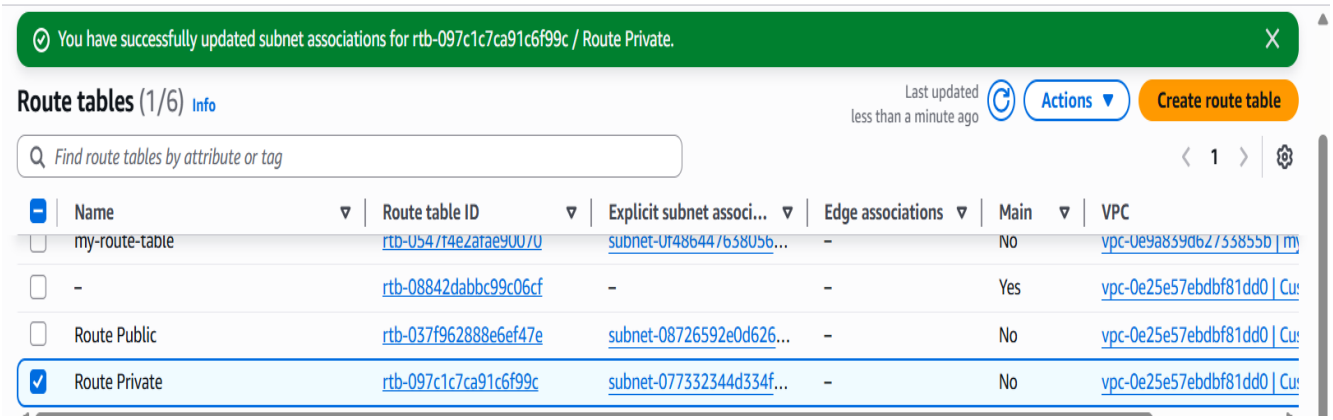
Created Route Table Private



Step-6 Updated the Internet Gateway for Route Public



Step-7 Subnet added for Private

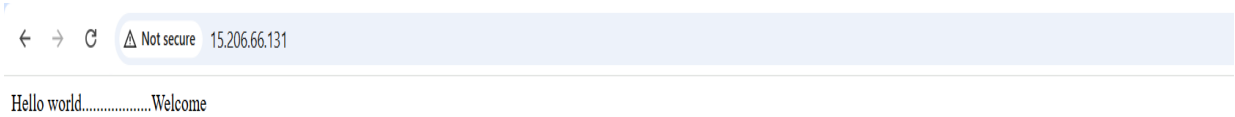


Step-8 Created 2 Instances named Public & Private

Step-9 Pinged the public instance

```
Last login: Thu Jun  4 06:13:16 2026 from 116.68.105.180
[ec2-user@ip-192-168-1-15 ~]$ sudo -i
[root@ip-192-168-1-15 ~]# ping google.com
PING google.com (142.251.222.78) 56(84) bytes of data.
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=1 ttl=117 time=1.80 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=2 ttl=117 time=1.82 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=3 ttl=117 time=1.83 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=4 ttl=117 time=1.83 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=5 ttl=117 time=1.82 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=6 ttl=117 time=1.82 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=7 ttl=117 time=1.82 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=8 ttl=117 time=1.82 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=9 ttl=117 time=1.83 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=10 ttl=117 time=1.83 ms
```

Step-10 Host a website inside public web server



Hello world.....Welcome

Step-11 Private instance Connected to Public using of ssh

```
[root@ip-192-168-1-15 ~]# vim newkey.pem
[root@ip-192-168-1-15 ~]# chmod 400 newkey.pem
[root@ip-192-168-1-15 ~]# ssh -i newkey.pem ec2-user@192.168.2.108
The authenticity of host '192.168.2.108 (192.168.2.108)' can't be established.
ED25519 key fingerprint is SHA256:pF9r0nwq8QjMOuZVen5I2PA9+cHZ9aKqQ5xWhxsZEQc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.2.108' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-192-168-2-108 ~]$
```

Step-12 Created Nat Gateway

NAT gateway nat-0f40377976a73b4ff | Natconnection was created successfully.

nat-0f40377976a73b4ff / Natconnection

Actions

Details

NAT gateway ID	Connectivity type	State	State message Info
nat-0f40377976a73b4ff	Public	Pending	-
NAT gateway ARN	**Primary public IPv4 address**	**Primary private IPv4 address**	**Primary network interface ID**
arn:aws:ec2:ap-south-1:280245841831:natgateway/nat-0f40377976a73b4ff	-	-	-
VPC	**Subnet**	**Created**	**Deleted**
vpc-0e25e57ebdbf81dd0 / Custom-VPC	subnet-08726592e0d6268a9 / Public Subnet	Thursday, 4 June 2026 at 13:10:30 GMT +5:30	-

Secondary IPv4 addresses | Monitoring | Tags

Step-13 NAT gateway attaches to private Route from Route table

Updated routes for rtb-097c1c7ca91c6f99c / Route Private successfully

Details

rtb-097c1c7ca91c6f99c / Route Private

Actions

Details Info

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-097c1c7ca91c6f99c	No	subnet-077332344d334fb29 / Private Subnet	-
VPC	**Owner ID**		
vpc-0e25e57ebdbf81dd0 | Custom-VPC	280245841831		

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0f40377976a73b4ff	Active	No	Create Route
192.168.0.0/16	local	Active	No	Create Route Table

Step-14 Pinged the private instance checking it having the internet connectivity

```
Last login: Thu Jun  4 07:05:34 2026 from 116.68.105.180
[ec2-user@ip-192-168-1-15 ~]$ sudo -i
[root@ip-192-168-1-15 ~]# ping google.com
PING google.com (142.250.143.139) 56(84) bytes of data.
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=1 ttl=113 time=1.71 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=2 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=3 ttl=113 time=1.68 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=4 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=5 ttl=113 time=1.69 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=6 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=7 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=8 ttl=113 time=1.69 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=9 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=10 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=11 ttl=113 time=1.68 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=12 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=13 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=14 ttl=113 time=2.32 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=15 ttl=113 time=1.65 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=16 ttl=113 time=1.68 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=17 ttl=113 time=1.74 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=18 ttl=113 time=1.80 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=19 ttl=113 time=1.66 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=20 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=21 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=22 ttl=113 time=1.67 ms
64 bytes from pt-in-f139.1e100.net (142.250.143.139): icmp_seq=23 ttl=113 time=1.67 ms
```

Step-15 Installed MariaDB server and enable the service in the private instance

```
[root@ip-192-168-2-108 ~]# yum install -y mariadb105-server
Last metadata expiration check: 0:20:51 ago on Thu Jun  4 09:08:34 2026.
Dependencies resolved.
================================================================================
 Package                    Arch   Version                     Repository   Size
================================================================================
Installing:
 mariadb105-server          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux  10 M
Installing dependencies:
 mariadb-connector-c        x86_64 3.3.10-1.amzn2023.0.1       amazonlinux 211 k
 mariadb-connector-c-config noarch 3.3.10-1.amzn2023.0.1       amazonlinux 9.9 k
 mariadb105                 x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 1.5 M
 mariadb105-common          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux  28 k
 mariadb105-errmsg          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 212 k
 mysql-selinux              noarch 1.0.4-2.amzn2023.0.3        amazonlinux  36 k
 perl-DBD-MariaDB           x86_64 1.22-1.amzn2023.0.5         amazonlinux 152 k
 perl-DBI                   x86_64 1.647-1.amzn2023.0.2        amazonlinux 804 k
 perl-File-Copy             noarch 2.34-477.amzn2023.0.8       amazonlinux  20 k
 perl-Math-BigInt           noarch 1:1.9998.39-2.amzn2023.0.2 amazonlinux 202 k
 perl-Math-BigRat           noarch 0.2624-500.amzn2023.0.2     amazonlinux  42 k
 perl-Math-Complex          noarch 1.59-477.amzn2023.0.8       amazonlinux  47 k
 perl-Sys-Hostname          x86_64 1.23-477.amzn2023.0.8       amazonlinux  17 k
Installing weak dependencies:
 mariadb105-backup          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 6.0 M
 mariadb105-cracklib-password-check
                            x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux  13 k
 mariadb105-gssapi-server   x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux  15 k
 mariadb105-server-utils    x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 207 k

Transaction Summary
================================================================================
Install  18 Packages

Total download size: 20 M
Installed size: 117 M
```

```
  Installing       : mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.   16/18
  Running scriptlet: mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64    17/18
  Installing       : mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64    17/18
  Running scriptlet: mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64    17/18
  Installing       : mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x   18/18
  Running scriptlet: mysql-selinux-1.0.4-2.amzn2023.0.3.noarch            18/18
  Running scriptlet: mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x   18/18
  Verifying        : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      1/18
  Verifying        : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.n    2/18
  Verifying        : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64            3/18
  Verifying        : mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64     4/18
  Verifying        : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64     5/18
  Verifying        : mariadb105-cracklib-password-check-3:10.5.29-1.amz    6/18
  Verifying        : mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64     7/18
  Verifying        : mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.    8/18
  Verifying        : mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64     9/18
  Verifying        : mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x   10/18
  Verifying        : mysql-selinux-1.0.4-2.amzn2023.0.3.noarch            11/18
  Verifying        : perl-DBD-MariaDB-1.22-1.amzn2023.0.5.x86_64          12/18
  Verifying        : perl-DBI-1.647-1.amzn2023.0.2.x86_64                 13/18
  Verifying        : perl-File-Copy-2.34-477.amzn2023.0.8.noarch          14/18
  Verifying        : perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch   15/18
  Verifying        : perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch      16/18
  Verifying        : perl-Math-Complex-1.59-477.amzn2023.0.8.noarch       17/18
  Verifying        : perl-Sys-Hostname-1.23-477.amzn2023.0.8.x86_64       18/18

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64
  mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-cracklib-password-check-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
  mysql-selinux-1.0.4-2.amzn2023.0.3.noarch
  perl-DBD-MariaDB-1.22-1.amzn2023.0.5.x86_64
  perl-DBI-1.647-1.amzn2023.0.2.x86_64
  perl-File-Copy-2.34-477.amzn2023.0.8.noarch
  perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch
  perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch
  perl-Math-Complex-1.59-477.amzn2023.0.8.noarch
  perl-Sys-Hostname-1.23-477.amzn2023.0.8.x86_64

Complete!
[root@ip-192-168-2-108 ~]# systemctl start mariadb
[root@ip-192-168-2-108 ~]#
```

Step-16 Checking the status MariaDB server

```
[root@ip-192-168-2-108 ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-06-04 09:30:09 UTC; 4min 49s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 28335 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
    Process: 28357 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
    Process: 28455 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 28441 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 1014)
     Memory: 66.6M
        CPU: 590ms
     CGroup: /system.slice/mariadb.service
             └─28441 /usr/libexec/mariadbd --basedir=/usr

Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: The second is mysql@localhost, it has no password either, but
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: you need to be the system 'mysql' user to connect.
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: After connecting you can set the password, if you would need to be
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: able to connect as any of these users with a password and without sudo
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: Please report any problems at https://mariadb.org/jira
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: The latest information about MariaDB is available at https://mariadb.org/.
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: Consider joining MariaDB's strong and vibrant community:
Jun 04 09:30:08 ip-192-168-2-108.ap-south-1.compute.internal mariadb-prepare-db-dir[28396]: https://mariadb.org/get-involved/
Jun 04 09:30:09 ip-192-168-2-108.ap-south-1.compute.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[root@ip-192-168-2-108 ~]#
```

To update package: yum update mariadb105-server